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SEMICONDUCTOR DEVICE

Abstract

A semiconductor device includes a first semiconductor layer, first transistors, and second transistors. The first semiconductor layer includes a first face and a second face and includes a concave region and a convex region on the first face. The first transistors each include a first gate dielectric film located in the convex region of the first semiconductor layer, a side gate dielectric film located on sidewalls of the convex region, a first gate electrode located on the first gate dielectric film, and a side gate electrode located on the side gate dielectric film and connected to the first 10 gate electrode. The second transistors each include a second gate dielectric film located on the convex region and being thinner than the first gate dielectric film, and a second gate electrode located on the second gate dielectric film.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2024-037387, filed on Mar. 11, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The embodiments of the present invention relate to a semiconductor device.

BACKGROUND

[0003] Semiconductor devices in which a plurality of transistors are formed on a semiconductor substrate are known.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 is a block diagram illustrating a configuration example of a semiconductor storage device according to a first embodiment;

[0005] FIG. 2 is a sectional view illustrating a configuration example of the semiconductor storage device according to the first embodiment;

[0006] FIG. 3 is a schematic plan view illustrating a stacked body;

[0007] FIG. 4 is a sectional view illustrating memory cells of a three-dimensional structure;

[0008] FIG. 5 is a sectional view illustrating the memory cells of a three-dimensional structure;

[0009] FIG. 6 is a plan view illustrating a configuration example of transistors constituting logic circuits of a CMOS chip according to the first embodiment;

[0010] FIG. 7 is a sectional view illustrating a configuration example of the transistors constituting the logic circuits of the CMOS chip according to the first embodiment;

[0011] FIG. 8 is a plan view illustrating a configuration example of transistors constituting word line switches of the CMOS chip according to the first embodiment;

[0012] FIG. 9 is a sectional view illustrating a configuration example of the transistors constituting the word line switches of the CMOS chip according to the first embodiment;

[0013] FIG. 10 is a sectional view illustrating an example of a manufacturing method of a transistor according to the first embodiment;

[0014] FIG. 11 is a sectional view illustrating an example of the manufacturing method of a transistor subsequent to FIG. 10;

[0015] FIG. 12 is a sectional view illustrating an example of the manufacturing method of a transistor subsequent to FIG. 11;

[0016] FIG. 13 is a sectional view illustrating an example of the manufacturing method of a transistor subsequent to FIG. 12;

[0017] FIG. 14 is a sectional view illustrating an example of the manufacturing method of a transistor subsequent to FIG. 13;

[0018] FIG. 15 is a sectional view illustrating an example of the manufacturing method of a transistor subsequent to FIG. 14; and

[0019] FIG. 16 is a sectional view illustrating a configuration example of transistors constituting word line switches according to a second embodiment.

DETAILED DESCRIPTION

[0020] In general, according to the embodiment, a semiconductor device includes a first semiconductor layer, first transistors, and second transistors. The first semiconductor layer includes a first face and a second face on the opposite side to the first face, and includes a concave region recessed from the first face toward the second face and a convex region protruded with respect to the concave region. The first transistors each include a first gate dielectric film located in the convex region of the first semiconductor layer, a side gate dielectric film located on sidewalls of the convex region, a first gate electrode located on the first gate dielectric film, and a side gate electrode located on the side gate dielectric film and connected to the first gate electrode. The second transistors each include a second gate dielectric film located on the convex region and being thinner than the first gate dielectric film, and a second gate electrode located on the second gate dielectric film. Hereinafter, devices of the present disclosure will be described with reference to the drawings.

[0021] The present invention is not limited to the embodiments. In the present specification and the drawings, elements identical to those described in the foregoing drawings are denoted by like reference characters and detailed explanations thereof are omitted as appropriate.

First Embodiment

[0022] FIG. 1 is a block diagram illustrating a configuration example of a semiconductor storage device according to a first embodiment. The semiconductor storage device is, for example, a semiconductor memory such as a NAND flash memory. The semiconductor storage device includes a memory cell array MA, and peripheral circuits PC functioning as control circuits that control the memory cell array MA.

[0023] The memory cell array MA includes a plurality of memory blocks MB. Each of these memory blocks MB includes a plurality of string units SU. Each of these string units SU includes a plurality of memory strings MS. One ends of the memory strings MS are connected to the peripheral circuits PC with corresponding bit lines BL. The other ends of the memory strings MS are connected to the peripheral circuits PC with a common source layer SL.

[0024] Each of the memory strings MS includes a drain-side selection transistor STD, a plurality of memory cells MC, and a source-side selection transistor STS connected in series between the corresponding bit line BL and the source layer SL. Hereinafter, the drain-side selection transistor STD and the source-side selection transistor STS are also referred to simply as “selection transistors STD and STS”, or the like.

[0025] Each of the memory cells MC is an FET (Field Effect Transistor) including a semiconductor layer functioning as a channel region, a gate dielectric film including a charge accumulating film, and a gate electrode. The threshold voltage of each memory cell MC varies according to the amount of charges in the charge accumulating film. A word line WL is connected to the gate electrode of each of the memory cells MC corresponding to one memory string MS. Each of these word lines WL is connected in common to all the memory strings MS in one memory block MB.

[0026] The memory cells MC are provided corresponding to intersections between the word lines WL and the bit lines BL, respectively, and data can be written into or read from a memory cell MC through the corresponding bit line BL by application of a write voltage or a read voltage between the corresponding word line WL and the bit line BL. Data can be erased from the memory cell MC by application of an erase voltage between the corresponding word line WL and the corresponding bit line BL.

[0027] The selection transistors STD and STS are FETs each including a semiconductor layer functioning as a channel region, a gate dielectric film, and a gate electrode. Selection gate lines SGD and SGS are connected to the gate electrodes of the selection transistors STD and STS, respectively. The drain-side selection gate line SGD is provided corresponding to each of the string

units SU and is connected in common to all the memory strings MS in one string unit SU. The source-side selection gate line SGS is connected in common to all the memory strings MS in the string units SU in one memory block MB. A structure in which the source-side selection transistor STS, the memory cells MC, and the drain-side selection transistor STD are connected in series is referred to as “memory string” or “string unit”.

[0028] The peripheral circuits PC include an operating voltage generator **121** that generates an operating voltage, an address decoder **122** that decodes address data, a block selection circuit **123** and a voltage selection circuit **124** that transfer the operating voltage to the memory cell array MA in accordance with an output signal from the address decoder **122**, a sense amplifier module **125** connected to the bit lines BL, and a sequencer **126** that controls these components.

[0029] The operating voltage generator **121** includes a plurality of operating voltage output terminals **131**. The operating voltage generator **121** includes, for example, a step-down circuit such as a regulator, and a step-up circuit such as a charge pump circuit. The operating voltage generator **121** sequentially generates a plurality of operating voltages to be applied to the bit lines BL, the source layer SL, the word lines WL, and the selection gate lines SGD and SGS at the time of a read operation, a write operation, and an erase operation for the memory cell array MA in accordance with a control signal from the sequencer **126**, and outputs the generated operating voltages to the operating voltage output terminals **131**. The operating voltages output from the operating voltage output terminals **131** are appropriately adjusted in accordance with a control signal from the sequencer **126**.

[0030] The address decoder **122** includes a plurality of block selection lines BLKSEL and a plurality of voltage selection lines **133**. For example, the address decoder **122** sequentially refers to address data in an address register in accordance with a control signal from the sequencer **126**, decodes the address data, brings a block drive transistor **135** and voltage selection transistors **137** corresponding to the address data to an ON-state, and brings other block drive transistors **135** and other voltage selection transistors **137** to an OFF-state. For example, in a case where the block drive transistors **135** and the voltage selection transistors **137** are N-channel transistors, the address decoder **122** brings the voltages of a block selection line BLKSEL and a voltage selection line **133** corresponding to the address data to an “H” state and brings other voltages to an “L” state. In a case where the block drive transistors **135** and the voltage selection transistors **137** are P-channel transistors, it suffices to apply the opposite voltages to the block selection lines BLKSEL and the voltage selection lines **133**.

[0031] In the illustrated example, the address decoder **122** includes one block selection line BLKSEL for one memory block MB. However, this configuration can be appropriately changed. For example, one block selection line BLKSEL may be provided for two or more memory blocks MB.

[0032] The block selection circuit **123** includes a plurality of block selectors **134** each corresponding to one memory block MB. Each of these block selectors **134** includes a plurality of the block drive transistors **135** corresponding to the word lines WL and the selection gate lines SGD and SGS, respectively. The block drive transistors **135** are, for example, high breakdown voltage (HV (High Voltage)) transistors. The drain electrode (or the source electrode) of each of the block drive transistors **135** is electrically connected to the corresponding word line WL or selection gate line SGD or SGS. Each of the source electrodes (or the drain electrodes) is electrically connected to the operating voltage output terminal **131** via a line CG and the voltage selection circuit **124**. The gate electrodes thereof are connected in common to the corresponding block selection line BLKSEL.

[0033] The block selection circuit **123** further includes a plurality of transistors (not illustrated). These transistors are high breakdown voltage (HV) transistors connected between the selection gate line SGD or SGS and a ground voltage supply terminal. These transistors bring the selection gate lines SGD and SGS included in non-selected memory blocks MB to induction with the ground

voltage supply terminal. The word lines WL included in the non-selected memory blocks MB are brought to a floating state.

[0034] The voltage selection circuit **124** includes a plurality of voltage selectors **136** corresponding to the word lines WL and the selection gate lines SGD and SGS. Each of these voltage selectors **136** includes a plurality of the voltage selection transistors **137**. The voltage selection transistors **137** are, for example, high breakdown voltage (HV) transistors. The drain terminals (or the source terminals) of the voltage selection transistors **137** are electrically connected to the corresponding word lines WL and selection gate lines SGD and SGS via the corresponding lines CG and the block selection circuit **123**, respectively. The source terminals (or the drain terminals) are electrically connected to the corresponding operating voltage output terminals **131**, respectively. The gate electrodes are connected to the corresponding voltage selection lines **133**, respectively.

[0035] The sense amplifier module **125** is connected to the bit lines BL. The sense amplifier module **125** includes, for example, a plurality of sense amplifier units corresponding to the bit lines BL. Each of the sense amplifier units includes a clamp transistor that charges a bit line BL on the basis of the voltage generated by the operating voltage generator **121**, a sense transistor that senses the voltage or current of the bit line BL, and a plurality of latch circuits that retain an output signal, data to be written, and the like of the sense transistor.

[0036] The sequencer **126** outputs a control signal to the operating voltage generator **121**, the address decoder **122**, and the sense amplifier module **125** according to an input command and the state of the semiconductor device. For example, the sequencer **126** sequentially refers to command data in a command register in accordance with a clock signal, decodes the command data, and outputs the decoded data to the operating voltage generator **121**, the address decoder **122**, and the sense amplifier module **125**.

[0037] FIG. **2** is a sectional view illustrating a configuration example of a semiconductor storage device **1** according to a first embodiment. Hereinafter, the stacking direction of a stacked body **20** is assumed as a Z direction. One direction intersecting with, for example, being orthogonal to the Z direction is assumed as a Y direction. One direction intersecting with, for example, being orthogonal to the Z direction and the Y direction is assumed as an X direction.

[0038] The semiconductor storage device **1** includes an array chip **2** having a memory cell array MA, and a CMOS (Complementary Metal-Oxide Semiconductor) chip **3** having the peripheral circuits PC such as a CMOS circuit. The array chip **2** and the CMOS chip **3** are bonded on a bonding face B1 and are electrically connected to each other with lines joined on the bonding face. FIG. **2** illustrates a state in which the array chip **2** is located on the CMOS chip **3**.

[0039] The CMOS chip **3** includes a substrate **30**, transistors **31**, **135**, and **137**, vias **32**, lines **33** and **34**, and an interlayer dielectric film **35**.

[0040] The substrate **30** is, for example, a semiconductor substrate such as a silicon substrate. The transistors **31**, **135**, and **137** are NMOS or PMOS transistors provided on the substrate **30**. The transistors **31** are low breakdown voltage (LV (Low voltage) or VLV (Very Low Voltage)) transistors and constitute, for example, a CMOS circuit that controls the memory cell array MA of the array chip **2**. The transistors **31** constitute logic circuits such as a sense amplifier, a row decoder, and a column decoder. Therefore, the transistors **31** are required to operate at a high speed. Meanwhile, the transistors **135** and **137** are used as word line switches (drivers) for applying voltages to the word lines WL. Therefore, the transistors **135** are high breakdown voltage (HV) transistors having a higher breakdown voltage than the transistors **31** and function as block drive transistors of the block selection circuit **123** in FIG. **1**. The transistors **137** are also high breakdown voltage transistors having a higher breakdown voltage than the transistors **31** and function as voltage selection transistors of the voltage selection circuit **124** in FIG. **1**. It is preferable that the transistors **135** and **137** are high in the breakdown voltage, low in the threshold voltage, and low in the on-resistance while high-speed operations are also required. Semiconductor elements such as a resistive element and a capacitive element, other than the transistors **31**, **135**, and **137** may be

formed on the substrate **30**.

[0041] Each of the vias **32** electrically connects between one of the transistors **31**, **135**, and **137** and one of the lines **33** or between one of the lines **33** and one of the lines **34**. The lines **33** and **34** constitute a multilayer wiring structure in the interlayer dielectric film **35**. The lines **34** are embedded in the interlayer dielectric film **35** and are exposed on the surface of the interlayer dielectric film **35** to be substantially flush with the surface. The lines **33** and **34** are electrically connected to the transistors **31** and the like. For example, a metal such as copper or tungsten is used as the vias **32**, and the lines **33** and **34**. The interlayer dielectric film **35** coats and protects the transistors **31**, **135**, and **137**, the vias **32**, and the lines **33** and **34**. For example, an insulating film such as a silicon dioxide film is used as the interlayer dielectric film **35**.

[0042] The array chip **2** includes the stacked body **20**, columnar bodies CL, slits ST (LI), a source layer SL, a metallic layer **40**, contact plugs CCw, contact plugs **29**, bonding pads **50**, and an interlayer dielectric film **25**.

[0043] The stacked body **20** is provided above the transistors **31**, **135**, and **137** and is positioned in the Z direction with respect to the substrate **30**. The stacked body **20** is configured by alternately stacking a plurality of electrode films **21** and a plurality of insulating films **22** along the Z direction. The stacked body **20** constitutes the memory cell array MA. For example, a conductive metal such as tungsten is used as the electrode films **21**. For example, an insulating film such as a silicon dioxide film is used as the insulating films **22**. The insulating films **22** insulate the electrode films **21** from each other. That is, the electrode films **21** are stacked in a mutually insulated state. The numbers of stacked layers of the electrode films **21** and the insulating films **22** can be freely selected. The insulating films **22** may be, for example, porous insulating films or air gaps.

[0044] Ones or pluralities of the electrode films **21** at the top end and the bottom end of the stacked body **20** in the Z direction function as source-side selection gates SGS and drain-side selection gates SGD, respectively. Electrode films **21** between the source-side selection gates SGS and the drain-side selection gates SGD function as word lines WL. The word lines WL are gate electrodes of the memory cells MC. The source-side selection gates SGS are gate electrodes of source-side selection transistors. The drain-side selection gates SGD are gate electrodes of drain-side selection transistors. The source-side selection gates SGS are provided in an upper region of the stacked body **20**. The drain-side selection gates SGD are provided in a lower region of the stacked body **20**. The upper region indicates a region of the stacked body **20** on a side far from the CMOS chip **3** (a side close to the metallic layer **40**), and the lower region indicates a region of the stacked body **20** on a side close to the CMOS chip **3**.

[0045] The semiconductor storage device **1** includes a plurality of memory cells MC connected in series between one of the source-side selection transistors STS controlled by the source-side selection gates SGS and one of the drain-side selection transistors STD controlled by the drain-side selection gates SGD to constitute each memory string. Each of the memory strings is connected to, for example, a bit line BL through a via **28**. The bit lines BL are lines **23** provided below the stacked body **20** and extending in the X direction (the depth direction of the drawing of FIG. 2). Therefore, the bit lines BL are hereinafter referred to also as "bit lines **23**".

[0046] A plurality of columnar bodies CL are provided in the stacked body **20**. The columnar bodies CL extend in the stacked body **20** to penetrate through the stacked body **20** in the stacking direction (the Z direction) of the stacked body and are each located from a via **28** connected to a bit line **23** to the source layer SL. An internal structure of the columnar bodies CL will be described later. In the present embodiment, each of the columnar bodies CL is formed in two tiers in the Z direction. However, there is no problem with the columnar bodies CL in one tier. Alternatively, each of the columnar bodies CL may be formed in three or more tiers.

[0047] Although not illustrated in FIG. 2, a plurality of slits ST (see FIG. 3) are provided in the stacked body **20**. The slits ST extend in the Y direction and penetrate through the stacked body **20** in the stacking direction (the Z direction) of the stacked body **20**. An insulating film such as a

silicon dioxide film is filled in each of the slits ST and the insulating film is configured in a plate shape. The slits ST electrically isolate the electrode films **21** of the stacked body **20**. It is alternatively possible that the inner wall of each of the slits ST is coated with an insulating film such as a silicon dioxide film and that a conducting material is further embedded in the inner side of the insulating film. In this case, the conducting material can also function as a source line reaching the source layer SL.

[0048] The source layer SL is provided on the stacked body **20**. The source layer SL is an example of a first semiconductor layer. The source layer SL is provided corresponding to the stacked body **20**. The source layer SL has a face F1 and a face F2 on the opposite side to the face F1. The stacked body **20** (the memory cell array MA) is located on the side of the face F1 of the source layer SL, and the metallic layer **40** is located on the side of the face F2 thereof. The metallic layer **40** includes a source line **41** and a power-supply line **42**. The source layer SL is connected in common to one ends of the columnar bodies CL and provides a plurality of columnar bodies CL in the same memory cell array MA with a common source potential. That is, the source layer SL functions as a common source electrode of the memory cell array MA. For example, a conductive material such as a doped polysilicon is used for the source layer SL. For example, a metallic material of a lower resistance than that of the source layer SL, such as copper, aluminum, or tungsten is used for the metallic layer **40**. A reference sign **2s** denotes a stepped portion of the electrode films **21** provided to connect each of the contact plugs CCw to an electrode film **21**. The stepped portion **2s** will be described later with reference to FIG. 3.

[0049] Meanwhile, the bonding pads **50** are provided in a region that is above the stacked body **20** and where the source layer SL is not located. The bonding pads **50** are connected to metallic wires or the like (not illustrated) and receive power supply or a signal from outside the semiconductor storage device **1**. The bonding pads **50** are provided so as to be connected to one ends of the contact plugs **29** in the Z direction. The bonding pads **50** are connected to the transistors **31**, **135**, and **137** of the CMOS chip **3** via the contact plugs **29**, the lines **24**, and the lines **34**. Accordingly, external power supplied from the bonding pads **50** are supplied to the transistors **31**, **135**, and **137**. Alternatively, a signal or power is supplied to the transistors **31**, **135**, and **137** or the memory cell array MA via the bonding pads **50**.

[0050] The contact plugs CCw are provided in a peripheral part of the stacked body **20** and extend in the Z direction in the interlayer dielectric film **25**. Each of the contact plugs CCw is electrically connected between the electrode film **21** (a word line WL) and the line **24**. The contact plugs CCw are provided at the stepped portions **2s** each formed like stairs at each of the ends of the stacked body **20** and are each electrically connected to an associated electrode film **21**. The contact plugs CCw are provided to transmit a word line voltage from the CMOS chip **3** to the associated electrode films **21**. For example, a metal such as copper or tungsten is used for the contact plugs CCw.

[0051] The contact plugs **29** are provided in the peripheral part of the stacked body **20** and extend in the Z direction in the interlayer dielectric film **25**. Each of the contact plugs **29** is a contact plug provided from the line **24** to the bonding pad **50**. The contact plugs **29** are simultaneously formed in the same process as that of the contact plugs CCw connected to the word lines WL.

[0052] Each of the contact plugs **29** is electrically connected between the bonding pad **50** and the line **24**. The contact plugs **29** are used to supply a power-supply voltage or a signal from the bonding pads **50** to the array chip **2** or the CMOS chip **3**. For example, a metal such as copper or tungsten is used for the contact plugs **29**. The power-supply voltage is, for example, a power-supply voltage VDD or a reference voltage (for example, a ground voltage) VSS lower than the power-supply voltage VDD. The signal may be a control signal from outside, or may be data to be written or read data.

[0053] In the present embodiment, the array chip **2** and the CMOS chip **3** are individually formed and are bonded on the bonding face B1. Therefore, the transistors **31**, **135**, and **137** are not provided

in the array chip **2**. The stacked body **20** (the memory cell array MA) is not provided in the CMOS chip **3**. The transistors **31**, **135**, and **137** and the stacked body **20** are both on the side of the face F1 of the source layer SL. The metallic layer **40** is located on the side of the face F2 of the source layer SL.

[0054] The vias **28**, the lines **23**, and the lines **24** are provided in the-Z direction of the stacked body **20**. The lines **23** and **24** are embedded in the interlayer dielectric film **25**. The lines **24** are exposed on the surface of the interlayer dielectric film **25** to be substantially flush with the surface. The lines **23** and **24** are electrically connected to semiconductor bodies **210** (see FIG. **4**) of the columnar bodies CL. For example, a metal such as copper or tungsten is used for the vias **28**, the lines **23**, and the lines **24**. The interlayer dielectric film **25** coats and protects the stacked body **20**, the vias **28**, the lines **23**, and the lines **24**. For example, an insulating film such as a silicon dioxide film is used as the interlayer dielectric film **25**.

[0055] The interlayer dielectric film **25** and the interlayer dielectric film **35** are bonded on the bonding face B1 and, associated therewith, the lines **24** and the lines **34** are joined on the bonding face B1 to be substantially flush therewith. Accordingly, the array chip **2** and the CMOS chip **3** are electrically connected to each other via the lines **24** and the lines **34**.

[0056] FIG. **3** is a schematic plan view illustrating the stacked body **20**. The stacked body **20** includes the stepped portions **2s** and the memory cell array MA. The stepped portions **2s** are located, for example, at the ends of the stacked body **20**. The memory cell array MA is sandwiched or surrounded by the stepped portions **2s**. The slits ST (LI) are provided from the stepped portion **2s** at one end of the stacked body **20** through the memory cell array MA to the stepped portion **2s** at the other end of the stacked body **20**. Slits SHE are provided at least on the memory cell array MA. The slits SHE are shallower in the Z direction than the slits ST (LI) and extend substantially in parallel to the slits ST (LI). The slits SHE electrically isolate the electrode films **21** for each of the drain-side selection gates SGD. The slits ST may be source lines LI electrically isolated from the electrode films **21** of the stacked body **20** while being electrically connected to the source layer SL. That is, the slits ST may be source lines LI electrically isolated from the electrode films **21** of the stacked body **20** constituting the memory cell array MA and electrically connected to the source layer SL.

[0057] A portion of the stacked body **20** sandwiched by two slits ST illustrated in FIG. **3** is a memory block MB. A memory block MB constitutes, for example, a minimum unit of data erasing. A slit SHE is provided in each memory block MB. The stacked body **20** between a slit ST and a slit SHE is referred to as “finger”. The drain-side selection gates SGD are divided for each finger. Accordingly, at the time of writing and reading data, one finger in a memory block MB can be brought to a selected state by the associated drain-side selection gate SGD.

[0058] FIGS. **4** and **5** are sectional views illustrating memory cells of a three-dimensional structure. Each of the columnar bodies CL is provided in a memory hole MH formed in the stacked body **20**. Each of the columnar bodies CL penetrates through the stacked body **20** along the Z direction from one end part of the stacked body **20** to be provided in the stacked body **20** and the source layer SL. Each of the columnar bodies CL includes the semiconductor body **210**, a memory film **220**, and a core layer **230**. Each columnar body CL includes the core layer **230** located at a central part thereof, the semiconductor body (a semiconductor member) **210** located around the core layer **230**, and the memory film **220** located around the semiconductor body **210**. The semiconductor body **210** extends in the stacked body **20** in the stacking direction (the Z direction). The semiconductor body **210** is electrically connected to the source layer SL. The memory film **220** is located between the semiconductor body **210** and the electrode films **21** and has charge capturing parts. A plurality of the columnar bodies CL each selected from each of the fingers are connected in common to one bit line **23** through the vias **28** in FIG. **2**. Each of the columnar bodies CL is provided, for example, in a region of the memory cell array MA.

[0059] As illustrated in FIG. **5**, the shape of each of the memory holes MH in an X-Y plane is, for

example, circular or elliptic. A block dielectric film **221a** constituting a part of the memory film **220** may be provided between each of the electrode films **21** and adjacent insulating films **22**. The block dielectric film **221a** is, for example, a silicon oxide or a metal oxide. One example of the metal oxide is an aluminum oxide. A barrier film **21b** may be provided between each of the electrode films **21** and adjacent insulating films **22** and between each of the electrode films **21** and the memory film **220**. The barrier film **21b** is, for example, a laminated film including titanium nitride and titanium, for example, in a case where the electrode films **21** are tungsten. The block dielectric film **221a** suppresses back tunneling of charges from the electrode films **21** to the memory film **220**. The barrier film **21b** improves adhesion between the electrode films **21** and the block dielectric film **221a**.

[0060] The shape of the semiconductor body **210** is, for example, a bottomed tube. For example, polysilicon is used as the semiconductor body **210**. The semiconductor body **210** is, for example, undoped silicon. The semiconductor body **210** may be p-type silicon. The semiconductor body **210** functions as channels of the drain-side selection transistors, the memory cells MC, and the source-side selection transistors. That is, a plurality of the memory cells MC each have a storage region between the semiconductor body **210** and an electrode film **21** functioning as a word line WL and are stacked in the Z direction. One ends of a plurality of the semiconductor bodies **210** in the same memory cell array MA are electrically connected in common to the source layer SL.

[0061] The memory film **220** includes, for example, a cover dielectric film **221**, a charge capturing film **222**, a tunnel dielectric film **223**, and the block dielectric film **221a**. A portion of the memory film **220** other than the block dielectric film **221a** is located between the inner wall of the memory hole MH and the semiconductor body **210**. The shape of the memory film **220** is, for example, tubular. Each of the charge capturing film **223** extends in the Z direction. film **222** and the tunnel dielectric

[0062] The cover dielectric film **221** is located between the insulating films **22** and the charge capturing film **222** and between the block dielectric film **221a** and the charge capturing film **222**. The cover dielectric film **221** includes, for example, a silicon oxide. The cover dielectric film **221** protects the charge capturing film **222** from being etched when sacrificial films (not illustrated) are replaced by the electrode films **21** (in a replacement process). In a case where the replacement process is not used to form the electrode films **21**, it is possible that the cover dielectric film **221** is not provided.

[0063] The charge capturing film **222** is located between the cover dielectric film **221** and the tunnel dielectric film **223**. The charge capturing film **222** includes, for example, a silicon nitride and has a trap site that traps charges in the film. Portions of the charge capturing film **222** sandwiched between the electrode films **21** functioning as the word lines WL and the semiconductor body **210** constitute the storage regions of the memory cells MC as the charge capturing parts. A threshold voltage of each of the memory cells MC varies according to whether there are charges in the associated charge capturing part or the amount of charges captured in the charge capturing part. This causes each of the memory cells MC to retain information.

[0064] The tunnel dielectric film **223** is located between the semiconductor body **210** and the charge capturing film **222**. The tunnel dielectric film **223** includes, for example, a silicon oxide, or a silicon oxide and a silicon nitride. The tunnel dielectric film **223** is a potential barrier between the semiconductor body **210** and the charge capturing film **222**. For example, when electrons are injected from the semiconductor body **210** to the charge capturing film **222** (a write operation) and when positive holes are injected from the semiconductor body **210** to the charge capturing film **222** (an erase operation), the electrons and the positive holes each pass through (tunnel) the potential barrier of the tunnel dielectric film **223**.

[0065] The core layer **230** fills the internal space of the tubular semiconductor body **210**. The shape of the core layer **230** is, for example, columnar. The core layer **230** includes, for example, a silicon oxide and is insulative.

[0066] FIG. 6 is a plan view illustrating a configuration example of the transistors **31** constituting logic circuits of the CMOS chip **3** according to the first embodiment. FIG. 7 is a sectional view illustrating the configuration example of the transistors **31** constituting the logic circuits of the CMOS chip **3** according to the first embodiment. FIG. 7 illustrates a cross section along a line 7-7 in FIG. 6. FIGS. 6 and 7 illustrate a configuration of one transistor **31**.

[0067] As illustrated in FIG. 6, convex regions Raa are surrounded by a concave region Rsti. The convex regions Raa are, for example, element formation regions (active areas). The concave region Rsti is, for example, a formation region of element isolation parts STI (Shallow Trench Isolation). The concave region Rsti is filled with an insulating material such as a silicon dioxide film around the convex regions Raa on which the transistors **31** are formed.

[0068] Each of the transistors **31** has a drain **D31** located on the convex region Raa on one side of a gate electrode **G31**, and has a source **S31** located on the convex region Raa on the other side of the gate electrode **G31**.

[0069] Contacts **CT** are electrically connected to the lines **33** and **34** through the vias **32** or the like.

[0070] The substrate **30** in FIG. 7 can be, for example, a semiconductor substrate such as a silicon substrate. The substrate **30** is one example of a semiconductor layer. The substrate **30** includes a first face **F11** on which control circuits including the transistors **31**, **135**, and **137**, and the like are provided, and a second face **F12** on the opposite side to the first face **F11**. The substrate **30** has the concave region Rsti on the side of the first face **F1**, which is recessed from the first face **F11** toward the second face **F12**. The concave region Rsti is, for example, filled with an insulating film such as a silicon dioxide film and constitutes shallow trench isolation parts **STI31**. The shallow trench isolation parts **STI31** electrically isolate adjacent transistors **31** from each other. The substrate **30** has the convex regions Raa protruded with respect to the concave region Rsti. The convex regions Raa are surrounded by the concave region Rsti and are defined by the concave region Rsti.

[0071] The transistors **31** are provided on the first faces **F11** of the convex regions Raa. Each of the transistors **31** includes a gate dielectric film **IN31** and a gate electrode **G31**.

[0072] The gate dielectric film **IN31** is provided on the first faces **F11** of the convex regions Raa. The film thickness of the gate dielectric film **IN31** is smaller than that of a gate dielectric film **IN135** of the transistors **135** illustrated in FIG. 9. For example, an insulating film such as a silicon dioxide film is used as the gate dielectric film **IN31**. The gate electrode **G31** is provided on the gate dielectric film **IN31**.

[0073] The gate electrode **G31** is constituted of, for example, a laminated film including a first gate electrode part **G31p1**, a second gate electrode part **G31p2**, and a third gate electrode part **G31m**. The first and second gate electrode parts **G31p1** and **G31p2** are, for example, a conducting material such as doped polysilicon including boron as impurities. For example, a conductive metallic material such as tungsten or copper is used for the third gate electrode part **G31m**.

[0074] FIG. 8 is a plan view illustrating a configuration example of the transistors **135** constituting word line switches of the CMOS chip **3** according to the first embodiment. FIG. 9 is a sectional view illustrating the configuration example of the transistors **135** constituting the word line switches of the CMOS chip **3** according to the first embodiment. FIG. 9 illustrates a cross section along a line 9-9 in FIG. 8. FIGS. 8 and 9 illustrate the configuration of four transistors **135**. The configuration of the transistors **137** can be the same as that of the transistors **135**. Therefore, explanations of the configuration of the transistors **137** are omitted here.

[0075] As illustrated in FIG. 8, the convex regions Raa are surrounded by the concave region Rsti. Each of the transistors **135** has a drain **D135** provided in the convex region Raa on one side of a gate electrode **G135**, and has a source **S135** provided in the convex region Raa on the other side of the gate electrode **G135**. In FIG. 8, a plurality of the transistors **135** share the source **S135**.

[0076] A reference voltage source **150** is formed in the concave region Rsti and is constituted of the same material as that of the gate electrodes **G135**. The reference voltage source **150** is electrically connected to the substrate **30**, is kept at a predetermined reference voltage (for example, a ground

voltage), and electrically isolates transistors from each other.

[0077] The substrate **30** in FIG. **9** is the same substrate as the substrate **30** in FIG. **7**. Therefore, the transistors **31**, **135**, and **137** are formed on the same substrate **30**.

[0078] The transistors **135** are located on the first faces **F11** of the convex regions **Raa**. Each of the transistors **135** includes a gate dielectric film **IN135**, a side gate dielectric film **IN135s**, the gate electrode **G135**, and side gate electrodes **G135s**.

[0079] The gate dielectric film **IN135** is located on the first faces **F11** of the convex regions **Raa**. The film thickness of the gate dielectric film **IN135** is larger than that of the gate dielectric film **IN31** of the transistors **31** described above. The film thickness of the gate dielectric film **IN31** is, for example, equal to or less than 10 nanometers (nm) and the film thickness of the gate dielectric film **IN135** is, for example, equal to or more than 30 nm. For example, an insulating film such as a silicon dioxide film is used as the gate dielectric film **IN135**.

[0080] The side gate dielectric film **IN135s** is located on side surfaces **F11s** of the convex regions **Raa**. That is, the side gate dielectric film **IN135s** is located in concave portions of the concave region **Rsti** and is located on shallow trench isolation parts **STI135**. The thickness of the side gate dielectric film **IN135s** is larger than that of the gate dielectric film **IN135**. Accordingly, the breakdown voltage can be kept high while the gate width of the transistors **135** is increased. For example, an insulating film such as a silicon dioxide film is used as the side gate dielectric film **IN135s** similarly to the gate dielectric film **IN135**.

[0081] The gate electrode **G135** is located on the gate dielectric film **IN135**. The gate electrode **G135** is constituted of, for example, a laminated film including a first gate electrode part **G135p1**, a second gate electrode part **G135p2**, and a third gate electrode part **G135m**. The first and second gate electrode parts **G135p1** and **G135p2** are in the same layers as the first and second gate electrode parts **G31p1** and **G31p2**, respectively, and include, for example, boron as impurities. The third gate electrode part **G135m** is in the same layer as the third gate electrode part **G31m** and, for example, a conductive metallic material such as tungsten or copper is used therefor. The third gate electrode part **G135m** does not need to be a metallic material and it suffices that the third gate electrode part **G135m** is a material lower in the resistance than the first and second gate electrode parts **G31p1** and **G31p2**.

[0082] While each having a laminated structure including three layers in the present embodiment, the gate electrodes **G31** and **G135** may each be a single conducting layer, or have a laminated structure including two layers, or four or more layers.

[0083] The side gate electrodes **G135s** are located on the side gate dielectric film **IN135s** and are connected to the gate electrode **G135** on the first face **F11**. The side gate electrodes **G135s** are configured in the same layer as that of the first gate electrode part **G135p1**.

[0084] The side gate electrodes **G135s** are located in the concave portions of the concave region **Rsti** and are each embedded between the side gate dielectric films **IN135s** on the shallow trench isolation parts **STI135**.

[0085] Each of the side gate electrodes **G135s** is constituted of the second gate electrode part **G135p2** in the laminated structures of the gate electrodes **G31** and **G135** in the present embodiment. However, each of the side gate electrodes **G135s** may be constituted of the third gate electrode part **G135m**. Each of the side gate electrodes **G135s** may have a laminated structure including two or more layers.

[0086] The contacts **CT** are electrically connected to the lines **33** or **34** through the vias **32** or the like. The contacts **CT** on the source side or the drain side are electrically connected to the word lines **WL** through the contact plugs **CCw** of the array chip **2**.

[0087] In the formation region of the transistors **135**, only bottom portions of the concave regions **Rsti** are filled with, for example, an insulating film such as a silicon dioxide film and constitute the shallow trench isolation parts **STI135**. Each of the shallow trench isolation parts **STI135** electrically isolates adjacent transistors **135** from each other. The transistors **31**, **135**, and **137** in

FIGS. 7 to 9 may be provided on the same substrate **30** and on the same convex regions Raa.

[0088] According to the present embodiment, the transistors **31** constituting the logic circuits of the peripheral circuits PC of the CMOS chip **3** are low breakdown voltage (LV or VLV) transistors that each have the first face F11 of the active area Raa as a channel region and does not have a channel region on the side surfaces of the active area Raa as illustrated in FIG. 7. Accordingly, the logic circuits of the peripheral circuits PC consume low power, can perform a high-speed operation, and are efficiently downscaled.

[0089] Meanwhile, the block drive transistors **135** of the block selection circuit **123** and the voltage selection transistors **137** of the voltage selection circuit **124** are high breakdown voltage (HV) transistors that each have both the first face F11 and the side surfaces F11s of the active area Raa as a channel region and have the gate dielectric film IN135 thicker than that of the transistors **31** as illustrated in FIG. 9. Therefore, the channel widths (the gate widths) of the transistors **135** and **137** functioning as the word line switches (drivers) of the peripheral circuits PC are widened, which decreases the on-resistance and can improve the current drive capability for the word lines WL. Since the side surfaces F11s of the active area Raa are used as the channel region, increase in the planar sizes (the dimensions in the gate width direction) of the transistors **135** and **137** can be suppressed. Further, in a case where the current drive capability is maintained constant, the planar sizes (the dimensions in the gate width direction) of the transistors **135** and **137** can be decreased. In a case where a plurality of the transistors **135** and **137** are arranged in parallel as the word line switches, the interval (pitch) between the transistors **135** or the interval (pitch) between the transistors **137** can be decreased while the current drive capabilities of the transistors **135** and **137** are maintained.

[0090] The thickness of the side gate dielectric film IN135s of the transistors **135** and **137** is larger than that of the gate dielectric film IN135. Accordingly, the breakdown voltages of the transistors **135** and **137** can be kept to be substantially the same as that of the high breakdown voltage transistors that do not have a channel region on the side surfaces F11s. As a result, the transistors **135** and **137** according to the present embodiment can provide both the low on-resistance and the high breakdown voltage while maintaining the threshold voltage.

Manufacturing Method of Transistors **31**, **135**, and **137**

[0091] A manufacturing method of the transistors **31**, **135**, and **137** is explained next.

[0092] FIGS. 10 to 15 are sectional views illustrating an example of the manufacturing method of the transistors **31**, **135**, and **137** according to the first embodiment. The transistors **31**, **135**, and **137** are formed on the same substrate **30**. FIGS. 10 to 15 illustrate a transistor **31** and a transistor **135** side by side and illustrations of the transistors **137** are omitted. The transistors **137** can be formed in the same manner as the transistors **135**.

[0093] The gate dielectric films IN31 and IN135 are formed on the first face F11 of the substrate **30**. The gate dielectric film IN135 is formed thicker than the gate dielectric film IN31. For example, after an insulating film such as a silicon dioxide film is formed on the entire first face F11, the insulating film in the formation region of the gate dielectric film IN31 is selectively removed using a lithography technique and an etching technique, and the insulating film such as a silicon dioxide film is subsequently formed again on the entire first face F11. In this way, a relatively thick insulating film can be formed in the formation region of the gate dielectric film IN135, and a relatively thin insulating film can be formed in the formation region of the gate dielectric film IN31.

[0094] Next, a material (for example, doped polysilicon) of the first gate electrode parts G31p1 and G135p1 is formed on the gate dielectric films IN31 and IN135, and a material (for example, a silicon nitride film) of a hard mask HM1 is formed on the material of the first gate electrode parts G31p1 and G135p1.

[0095] Next, the material of the hard mask HM1 in the concave region Rsti is selectively etched using a lithography technique and an etching technique. The material of the first gate electrode

parts **G31p1** and **G135p1**, the gate dielectric films **IN31** and **IN135**, and the substrate **30** are processed using the hard mask **HM1** as a mask. Accordingly, trenches are formed in the concave region **Rsti**.

[0096] Next, a material (for example, a silicon dioxide film) of the shallow trench isolation parts **STI31** and **STI135** is formed in the trenches of the concave region **Rsti**. The material of the shallow trench isolation parts **STI31** and **STI135** is flattened using a CMP (Chemical Mechanical Polishing) method or the like. A structure illustrated in FIG. **10** is thereby obtained.

[0097] Next, the material of the shallow trench isolation parts **STI135** is selectively etched using a lithography technique and an etching technique. As illustrated in FIG. **11**, in the formation region of the transistors **135**, an upper portion of the material of the shallow trench isolation parts **STI135** in the concave region **Rsti** is removed to cause the shallow trench isolation parts **STI135** to remain at the bottom portions of the concave region **Rsti**. Accordingly, the side surfaces **F11s** of the active area **Raa** (portions of the sidewall of the concave region **Rsti**) are exposed in the formation region of the transistors **135**. The shallow trench isolation parts **STI135** in the formation region of the transistors **135** function as element isolation regions of the transistors **135**. Meanwhile, in the formation region of the transistors **31**, the concave region **Rsti** is kept filled with the material of the shallow trench isolation parts **STI31** as illustrated in FIG. **11**. The shallow trench isolation parts **STI31** in the formation region of the transistors **31** function as element isolation regions of the transistors **31**.

[0098] Next, a material (for example, a silicon dioxide film) of the side gate dielectric film **IN135s** is formed on the side surfaces **F11s** of the active areas **Raa** and the hard mask **MH1**. Next, the material of the side gate dielectric film **IN135s** is etched back. Accordingly, the side gate dielectric film **IN135s** is left on the side surfaces **F11s** of the active areas **Raa** in the formation region of the transistors **135** as illustrated in FIG. **12**. The structure in the formation region of the transistors **31** in FIG. **12** is unchanged from that in FIG. **11**.

[0099] Next, the hard mask **HM1** is removed using a hot phosphoric acid solution or the like as illustrated in FIG. **13**.

[0100] Next, a material (for example, doped polysilicon) of the gate electrode parts **G31p2** and **G135p2** is deposited on the shallow trench isolation parts **STI31** and **STI135**, the side gate dielectric film **IN135s**, and the gate electrode parts **G31p1** and **G135p1** as illustrated in FIG. **14**. Accordingly, the gate electrode part **G135p2** is embedded in the inner side of the concave region **Rsti** in the formation region of the transistors **135**. The gate electrode part **G135p2** opposes the side surfaces **F11s** of the active areas **Raa** with the side gate dielectric film **IN135s** interposed therebetween, and contributes to increase in the channel width (the gate width) of the transistors **135**.

[0101] Next, an upper portion of the material of the gate electrode parts **G31p2** and **G135p2** is etched back. Next, as illustrated in FIG. **15**, a material (for example, tungsten) of the third gate electrode parts **G31m** and **G135m** is deposited on the gate electrode parts **G31p2** and **G135p2**. The gate electrodes **G31** and **G135** are thereby formed.

[0102] Next, an interlayer dielectric film, via contacts, a wiring structure, and the like are formed on the gate electrodes **G31** and **G135** to complete the CMOS chip **3** although not illustrated.

[0103] Subsequently, the CMOS chip **3** and the array chip **2** formed separately from the CMOS chip **3** are bonded to each other, whereby the semiconductor storage device **1** illustrated in FIG. **2** is completed.

[0104] As described above, according to the present embodiment, the transistors **135** and **137** can be simultaneously formed on the same substrate as that for the transistors **31** in common semiconductor manufacturing processes. Accordingly, fewer additional manufacturing processes are required and increase in the manufacturing cost is suppressed.

Second Embodiment

[0105] FIG. **16** is a sectional view illustrating a configuration example of the transistors **135**

constituting word line switches according to a second embodiment. The configuration of the transistor **137** may be the same as that of the transistor **135**. Therefore, explanations of the configuration of the transistor **137** will be omitted.

[0106] In the second embodiment, the material (for example, doped polysilicon) of the side gate electrodes **G135s** in the concave region **Rsti** of the transistors **135** does not fill the concave region **Rsti** while coating the side gate dielectric film **IN135s**.

[0107] Meanwhile, the material (for example, tungsten) of the third gate electrode part **G135m** coats over the side gate electrodes **G135s** in the concave region **Rsti**. The material of the third gate electrode part **G135m** may be filled in the concave region **Rsti**. Even with this configuration, the transistor **135** according to the second embodiment has functions identical to those of the transistor **135** according to the first embodiment.

[0108] Other configurations of the second embodiment may be identical to those of the first embodiment. Therefore, the second embodiment can attain effects identical to those of the first embodiment.

[0109] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A semiconductor device comprising: a first semiconductor layer comprising a first face and a second face on an opposite side to the first face, and comprising a concave region recessed from the first face toward the second face and a convex region protruded with respect to the concave region; first transistors each comprising a first gate dielectric film located in the convex region of the first semiconductor layer, a side gate dielectric film located on sidewalls of the convex region, a first gate electrode located on the first gate dielectric film, and a side gate electrode located on the side gate dielectric film and connected to the first gate electrode; and second transistors each comprising a second gate dielectric film located on the convex region and being thinner than the first gate dielectric film, and a second gate electrode located on the second gate dielectric film.
2. The device of claim 1, wherein the side gate dielectric film is thicker than the first gate dielectric film.
3. The device of claim 1, wherein each of the first and second gate electrodes is constituted of a laminated film including a first conducting layer and a second conducting layer located on the first face of the convex region, and the side gate electrode is constituted of the second conducting layer located on the side gate dielectric film.
4. The device of claim 1, wherein the first and second transistors are located in the same convex region.
5. The device of claim 1, further comprising a memory cell array comprising a plurality of memory cells each provided corresponding to one of intersections between first lines and second lines and each enabling data to be written therein or read therefrom through an associated one of the second lines by application of a voltage to an associated one of the first lines, wherein sources or drains of the first transistors are electrically connected to the first lines.
6. The device of claim 5, wherein a first chip comprising the first transistors, the second transistors, and the first semiconductor layer and a second chip comprising the memory cell array are bonded to each other.
7. The device of claim 1, wherein an insulating material as well as the side gate dielectric film and

the side gate electrode is filled in the concave region.

8. A semiconductor device comprising: a first semiconductor layer comprising a first face and a second face on an opposite side to the first face, and comprising a concave region recessed from the first face toward the second face and a convex region protruded with respect to the concave portion; and first transistors each comprising a first gate dielectric film located in the convex region of the first semiconductor layer, a side gate dielectric film located on sidewalls of the convex region and being thicker than the first gate dielectric film, a first gate electrode located on the first gate dielectric film, and a side gate electrode located on the side gate dielectric film and connected to the first gate electrode.

9. The device of claim 8, wherein the first gate electrode is constituted of a laminated film including a first conducting layer and a second conducting layer located on the first face of the convex region, and the side gate electrode is constituted of the second conducting layer located on the side gate dielectric film.

10. The device of claim 8, further comprising second transistors each comprising a second gate dielectric film located on the convex region and being thinner than the first gate dielectric film, and a second gate electrode located on the second gate dielectric film, wherein the first and second transistors are located in the same convex region.

11. The device of claim 10, further comprising a memory cell array comprising a plurality of memory cells each provided corresponding to one of intersections between first lines and second lines and each enabling data to be written therein or read therefrom through an associated one of the second lines by application of a voltage to an associated one of the first lines, wherein sources or drains of the first transistors are electrically connected to the first lines.

12. The device of claim 11, wherein a first chip comprising the first transistors, the second transistors, and the first semiconductor layer and a second chip comprising the memory cell array are bonded to each other.

13. The device of claim 8, wherein an insulating material as well as the side gate dielectric film and the side gate electrode is filled in the concave region.

14. A semiconductor device comprising: a memory cell array comprising a plurality of memory cells each provided corresponding to one of intersections between first lines and second lines and each enabling data to be written therein or read therefrom through an associated one of the second lines by application of a voltage to an associated one of the first lines; a first semiconductor layer comprising a first face and a second face on an opposite side to the first face, and comprising a concave region recessed from the first face toward the second face and a convex region protruded with respect to the concave region; and first transistors each comprising a first gate dielectric film located in the convex region of the first semiconductor layer, a side gate dielectric film located on sidewalls of the convex region, a first gate electrode located on the first gate dielectric film, and a side gate electrode located on the side gate dielectric film and connected to the first gate electrode, wherein sources or drains of the first transistors are electrically connected to the first lines.

15. The device of claim 14, wherein the side gate dielectric film is thicker than the first gate dielectric film.

16. The device of claim 15, wherein each of the first and second gate electrodes is constituted of a laminated film including a first conducting layer and a second conducting layer located on the first face of the convex region, and the side gate electrode is constituted of the second conducting layer located on the side gate dielectric film.

17. The device of claim 14, further comprising second transistors each comprising a second gate dielectric film located on the convex region and being thinner than the first gate dielectric film, and a second gate electrode located on the second gate dielectric film, wherein the first and second transistors are located in the same convex region.

18. The device of claim 17, wherein a first chip comprising the first transistors, the second transistors, and the first semiconductor layer and a second chip comprising the memory cell array

are bonded to each other.

19. The device of claim 14, wherein an insulating material as well as the side gate dielectric film and the side gate electrode is filled in the concave region.
